

Efficacy and Safety of Fibroblast Growth Factor-21 Analogs for the Treatment of Metabolic Dysfunction-Associated Steatohepatitis: A Systematic Review and Meta-Analysis

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Fibroblast growth factor (FGF)-21 analogs are potential therapeutic candidates for metabolic dysfunction-associated steatohepatitis (MASH). This systematic review and meta-analysis aimed to assess the efficacy and safety of the FGF-21 analogs, efruxifermin, pegbelfermin, and pegozafermin for MASH treatment. A comprehensive systematic review and meta-analysis of randomized controlled trials from five major databases was conducted. Primary efficacy outcomes focused on liver histological improvement, while secondary efficacy outcomes encompassed reductions in liver fat content and improvements in biochemical parameters. Safety outcomes examined included treatment-emergent adverse events (TEAEs), treatment-related TEAEs, TEAEs leading to discontinuation, and serious TEAEs. Eight eligible studies involving 963 patients were included in this review. Compared with the placebo group, the FGF-21 analog-treated group exhibited significantly improved primary efficacy outcomes, specifically ≥ 1 stage improvement in fibrosis with no worsening of MASH (risk ratio [RR]=1.83; 95% confidence interval [CI]=1.27–2.62) and at least two-point improvement in the non-alcoholic fatty liver disease activity score with no worsening of fibrosis (RR=2.85; 95% CI=2.06–3.95). Despite an increased risk of TEAEs (RR=1.17; 95% CI=1.08–1.27) and treatment-related adverse events (RR=1.75; 95% CI=1.40–2.19), FGF-21 analogs exhibited an acceptable safety profile. FGF-21 analogs were significantly better in achieving liver histological improvements and beneficial biochemical outcomes compared with placebo, with a tolerable safety pattern. These findings shed light on the efficacy and safety of FGF-21 analogs and provide valuable evidence for their application as MASH therapeutics.

Study Highlights

WHAT IS THE CURRENT KNOWLEDGE ON THE TOPIC?

✓ FGF-21 governs lipid and glucose metabolism, effecting a reduction in lipogenesis and an enhancement in hepatic insulin sensitivity, ultimately ameliorating the lipid profile. Three FGF-21 analogs have undergone clinical trials for MASH treatment.

WHAT QUESTION DID THIS STUDY ADDRESS?

✓ This systematic review and meta-analysis aimed to evaluate the efficacy and safety of FGF-21 analogs (efruxifermin, pegbelfermin, and pegozafermin) in treating MASH.

WHAT DOES THIS STUDY ADD TO OUR KNOWLEDGE?

✓ This study represents the first comprehensive systematic review and meta-analysis demonstrating the efficacy and

safety of FGF-21 analogs in MASLD/MASH patients. FGF-21 analogs exhibited superior liver histological improvements and beneficial biochemical outcomes compared with placebo, alongside an acceptable safety profile. Notably, variations in efficacy were observed among analogs, with pegozafermin showing the highest effectiveness.

HOW MIGHT THIS CHANGE CLINICAL PHARMACOLOGY OR TRANSLATIONAL SCIENCE?

✓ These findings shed light on the efficacy and safety profiles of FGF-21 analogs, providing evidence for their potential application as MASLD/MASH therapeutics.

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Metabolic dysfunction-associated steatotic liver disease (MASLD, formerly non-alcoholic fatty liver disease, NAFLD) encompasses various pathologies from uncomplicated hepatic fat accumulation (simple steatosis) to lobular inflammation and hepatocyte ballooning, known as metabolic dysfunction-associated steatohepatitis (MASH, formerly known as non-alcoholic steatohepatitis, NASH).^{1,2} Hepatic steatosis is caused by decreased mitochondrial fatty acid oxidation, increased hepatic lipogenesis, and reduced lipid export from hepatocytes.³ Major causes of MASLD include insulin resistance, dysfunctional lipid metabolism, and inflammation.⁴ MASLD is estimated to affect one in four individuals worldwide, and patients with MASLD, especially those with cirrhosis, exhibit an increased risk of hepatocellular carcinoma.⁵

Although MASLD is the most common liver disease worldwide with a global prevalence of ~30%,⁶ no pharmacotherapy is available for MASLD treatment. Currently, MASLD is managed via dietary and lifestyle modifications. Practice guidelines endorse the off-label use of vitamin E for non-cirrhotic and non-diabetic patients with biopsy-proven MASH and pioglitazone for patients with biopsy-proven MASH.¹ Rapid increase in the global prevalence of MASLD necessitates the development of novel treatments. The lack of available drugs for MASLD may be due to several factors, including the requirement for multifunctional agents and insufficient knowledge on key factors and appropriate drug targets for MASLD.⁷ Slow and long-term progression of the disease also poses a significant challenge to the conduct of clinical trials.

Despite these challenges, pharmacological approaches targeting various cellular components or molecular pathways of MASLD/MASH have been developed based on the pathophysiological mechanisms of MASLD.⁸ Numerous targets have been explored for MASLD/MASH treatment, including fibroblast growth factor (FGF), thyroid hormone receptors, glucagon-like peptide 1, peroxisome proliferator-activated receptors, and agents influencing hepatic steatosis synthesis, inflammatory pathways, genetic components, and fibrosis mechanisms.⁹ Presently, ~40 compounds are advancing in clinical trials, reflecting extensive research and exploration of various drug mechanisms.¹⁰ Recently, the Food and Drug Administration (FDA) granted accelerated approval to resmetirom (a thyroid hormone receptor- β agonist), and efforts are ongoing to develop other favorable targets for MASLD/MASH treatment.

Among these targets, FGF-21 stands out as a promising candidate for MASLD treatment. FGF-21 regulates lipid and glucose metabolism, playing pivotal roles in reducing lipogenesis and enhancing hepatic insulin sensitivity, thereby improving the lipid profile.¹¹ Previous preclinical studies have consistently demonstrated the potent therapeutic benefits of FGF-21, not only for MASLD but also for related conditions such as obesity, insulin resistance, hyperglycemia, dyslipidemia, atherosclerosis, and diabetic cardiomyopathy.¹² However, use of endogenous FGF-21 is limited by its short half-life, poor stability, and poor bioavailability.¹³ To

overcome these challenges, FGF-21 analogs are being developed as novel treatments for MASLD/MASH.

To date, three FGF-21 analogs have been tested in clinical trials.⁹ The first candidate, pegbelfermin (BMS-986036), was developed by Bristol Myers Squibb. However, further development of this drug was terminated due to concerns about its clinical efficacy in phase IIb trials.¹⁴ Efruxifermin (AR001) and pegozafermin (BIO89-100) are subcutaneous long-acting Fc-FGF-21 fusion proteins developed by Akero Therapeutics and 89-BIO, respectively. Quantitative clinical efficacy and safety data are available for each FGF-21 analog, based on randomized controlled trials (RCTs). However, a systematic review and meta-analysis with a sufficient sample size is necessary to comprehensively determine the efficacy and safety of these analogs. To the best of our knowledge, no systematic review or meta-analysis has examined the clinical efficacy and safety of the pegbelfermin, efruxifermin, and pegozafermin FGF-21 analogs to date.

Therefore, in this study, we aimed to conduct a systematic review and meta-analysis of relevant RCTs to assess the clinical efficacy and safety of these three FGF-21 analogs compared with placebo for MASLD/MASH treatment.

METHODS

This study was performed in accordance with the Cochrane Methodology and Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.¹⁵ It has been registered on PROSPERO under the identifier CRD42023451595.

Literature search

The ClinicalTrials.gov, Cochrane Library, Cumulative Index to Nursing and Allied Health Literature, PubMed, and World Health Organization International Clinical Trials Registry Platform databases were comprehensively searched from inception to August 29, 2023. The search query was as follows: (((Non-alcoholic fatty liver disease OR Non-alcoholic steatohepatitis) OR (Non-alcoholic fatty liver disease OR NAFLD OR Fatty liver, Non-alcoholic OR Liver, Non-alcoholic fatty OR Non-alcoholic steatohepatitis OR NASH)) AND ((Fibroblast growth factor-21 OR fgf-21 OR Fibroblast growth factor21 OR fgf21) OR (Fibroblast growth factor-21 analog* OR fgf-21 analog* OR Fibroblast growth factor21 analog* OR fgf21 analog* OR Efruxifermin OR Pegbelfermin OR Pegozafermin OR AKR-001 OR BMS-986036 OR BIO89-100))). The search terms for each database are listed in **Table S1**.

Study inclusion criteria and selection

The search was confined to RCTs published in English, with no restrictions on the year of publication, number of participants, study duration, or country. Articles meeting the following criteria were included: (i) population: adult patients with a definitive diagnosis of MASLD or MASH established via liver biopsy or imaging and (ii) intervention: examination of FGF-21 analog treatment. Exclusion criteria included trials without any patients with MASLD/MASH, open-label trials, those with incomplete data, and those involving combination drug therapy as well as review articles and *post hoc* analyses.

Bibliographic citations were imported into EndNote 20 (Clarivate, Philadelphia, PA) and duplicates were removed. All references were exported to a Microsoft Excel spreadsheet for study selection and data extraction. Two independent investigators screened the studies to determine their eligibility for inclusion. The titles and abstracts of the identified studies were screened in the first stage, and the full text of each article shortlisted for inclusion was reviewed in the second stage. Any disagreements between the investigators regarding the eligibility of the study were resolved via discussion. In cases of continued disagreement, a third party was consulted for the final decision.

Data extraction and outcome measures

Data extraction was conducted by one investigator using a specifically designed form, and a second investigator cross-checked all the extracted data to ensure accuracy. Discrepancies were resolved via discussion. The following information was systematically collected from each study: study title, clinical trial number, phase of clinical trial, study period, type and dose of the FGF-21 analog used, type of control, sample size, participant demographic data, and efficacy and safety outcomes.

Primary efficacy outcomes in the meta-analysis included liver histological improvement, such as achieving a ≥ 1 stage improvement in fibrosis with no worsening of MASH, at least a two-point improvement in the MASLD activity score (NAS) with no worsening of fibrosis, and MASH resolution with no worsening of fibrosis. Secondary efficacy outcomes included a reduction in the liver fat content (LFC) and improvements in various biochemical outcomes. LFC reduction was defined as a relative decrease of at least 30% in the magnetic resonance imaging-proton density fat fraction from baseline. Improvement in biochemical outcomes was based on percentage change from baseline in blood and serum levels of alanine aminotransferase (ALT), aspartate aminotransferase (AST), triglycerides (TG), low-density lipoprotein cholesterol (LDL-C), high-density lipoprotein cholesterol (HDL-C), adiponectin, and N-terminal type III collagen pro-peptide (Pro-C3; serum markers of fibrosis and fibrogenesis).

Safety outcomes examined in this meta-analysis included treatment-emergent adverse events (TEAEs), treatment-related TEAEs, TEAEs leading to discontinuation, serious TEAEs, injection site erythema, diarrhea, and nausea.

Risk of bias (ROB) assessment

Two independent reviewers used the Cochrane Handbook ROB assessment tool to evaluate the ROB in each included study. Discrepancies were resolved via discussion, and if necessary, additional consultation was sought. The methodological aspects assessed for ROB included random sequence generation and allocation concealment (selection bias), blinding of participants and personnel (performance bias), blinding of outcome assessment (detection bias), incomplete outcome data (attrition bias), selective reporting (reporting bias), and other biases. The quality assessment results of the ROB tool for each study were visualized using the Review Manager 5.4 software (RevMan; The Nordic Cochrane Center, The Cochrane Collaboration, Copenhagen, Denmark).

Data analysis

All meta-analyses were conducted using the RevMan software. For dichotomous variables, the effect denotes the risk ratio (RR) with a 95% confidence interval (CI) using the Mantel-Haenszel method. For continuous variables, the mean difference (MD) and 95% CI were calculated between the groups using the inverse-variance model. Statistical tests were two-sided, and P values ≤ 0.05 were considered statistically significant. In the case of the presence of subgroups in the same arm, estimates were combined to obtain an overall value according to Cochrane guidelines.¹⁶ A Chi-square (χ^2) test was used to

evaluate the statistical heterogeneity among the studies, and I^2 statistics were used to quantify the degree of heterogeneity. Heterogeneity was categorized as follows: $I^2 = 0$ –40% (no significant heterogeneity), 30–60% (moderate heterogeneity), 50–90% (substantial heterogeneity), and 75–100% (considerable heterogeneity).¹⁷ If the I^2 statistic was $>50\%$, a random effects model was used. If no considerable heterogeneity was apparent among the studies, a fixed-effects model was used. Subgroup analysis was conducted based on the use of different FGF-21 analogs in the treatment group, and sensitivity analysis was performed by removing each study separately. Publication biases were visually assessed using funnel plots and statistically examined using Begg's and Egger's tests.

RESULTS

Study selection

A total of 586 studies were initially identified by searching through five databases. After removing 120 duplicates, 466 unique studies were obtained. Subsequently, screening of titles and abstracts led to the assessment of 34 full-text articles for eligibility. Among these, 26 articles were excluded based on predefined criteria: trials without MASLD/MASH patients ($n = 7$), incomplete data ($n = 3$), open-label trials ($n = 4$), review articles/*post hoc* analyses ($n = 11$), and combination drug therapy studies ($n = 1$). Ultimately, eight RCTs, including two studies (three articles) on efruxifermin,^{18–20} three studies (three articles) on pegbelfermin,^{14,21,22} and two studies (two articles) on pegozafermin,^{23,24} were selected for the present systematic review and meta-analysis. A PRISMA flowchart depicting the search process, screening, and study selection criteria is shown in Figure 1.

Study characteristics and ROB assessment

Table 1 presents the key characteristics of the studies and associated patients included in this systematic review and meta-analysis. The included study publication dates spanned from 2019 to 2023, with four studies classified as phase IIa trials and four as phase IIb trials. Cumulative participation across these studies involved 963 patients, with 685 receiving FGF-21 analogs and 278 receiving a placebo. FGF-21 analogs were administered at various doses: 28–70 mg once per week for efruxifermin, 10–50 mg once per week for pegbelfermin, and 18–36 mg once per week or biweekly for pegozafermin. Among all participants, 591 (61.4%) were female and 752 (78.1%) were white. In addition, 605 patients (62.8%) had type 2 diabetes as a comorbidity. Further details regarding the baseline measurements of LFC, NAS, and fibrosis stages of the patients are summarized in Table S2.

ROB assessment results for all included trials are shown in Figure S1. Overall, the studies were deemed to have a low ROB in random sequence generation, blinding of participants and personnel, blinding of outcome assessments, and selective reporting. However, three articles were noted to have a high risk of incomplete outcome data bias, as they did not provide a rational methodology or information about the treatment for missing data or withdrawn patients. Two other studies were judged to have an unclear risk of incomplete outcome data bias due to insufficient information. One study involving efruxifermin

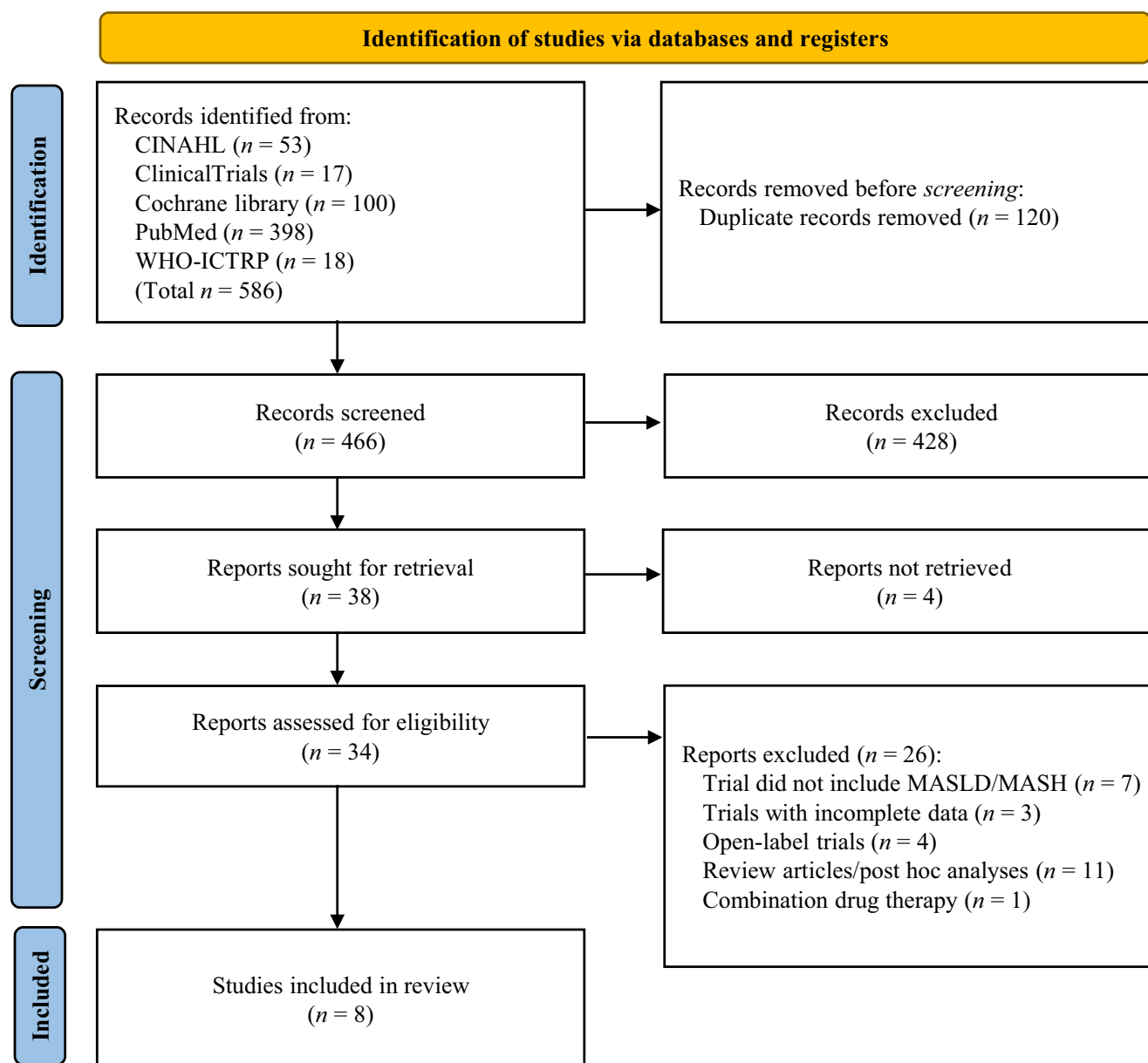


Figure 1 Flow diagram of the search strategy and study selection criteria based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.

(NCT04767529) was presented in abstract form only, resulting in all biases associated with this trial being judged as “unclear” or “high” risk overall.

Liver histological improvements

FGF-21 analogs significantly improved the primary efficacy outcomes, specifically a ≥ 1 stage improvement in fibrosis with no worsening of MASH and at least a two-point improvement in the NAS with no worsening of fibrosis, compared with placebo. Analysis of primary efficacy outcomes related to liver histological improvements incorporated data from six RCTs involving 715 patients (518 receiving FGF-21 analogs and 197 receiving a placebo).

The pooled estimate for achieving a ≥ 1 stage improvement in fibrosis with no worsening of MASH was 1.83 (95%

CI = 1.27–2.62; $P = 0.001$), with moderate heterogeneity ($\chi^2 = 9.58$; $I^2 = 48\%$; $P = 0.09$) among the studies (Figure 2). In subgroup analysis, pegbelfermin showed the lowest efficacy (RR = 1.13; 95% CI = 0.69–1.86; $P = 0.62$), whereas pegzofermin exhibited the highest efficacy (RR = 3.96; 95% CI = 1.47–10.66; $P = 0.0006$) in achieving a ≥ 1 stage improvement in fibrosis with no worsening of MASH, even though it was based on a single trial.

The pooled estimate for at least a two-point improvement in NAS with no worsening of fibrosis was 2.85 (95% CI = 2.06–3.95; $P < 0.00001$), with low heterogeneity ($\chi^2 = 3.33$; $I^2 = 0\%$; $P = 0.65$) among the studies (Figure 3). In subgroup analysis, pegzofermin showed the lowest efficacy (RR = 2.48; 95% CI = 1.57–3.93; $P = 0.0001$), whereas efruxifermin exhibited the highest efficacy

Table 1 Key characteristics of the studies and associated patients included in the systematic review and meta-analysis

Drug (ClinicalTrials #)	Publication year	Phase	Study period	Intervention			Age (Mean [SD], year)		Female (n [%])		White (n [%])		BMI (Mean [SD], kg/m ²)		T2D (n [%])	
				FGF-21 analogs (n)	Dose (n)	Control (n)	FGF-21 analogs	Placebo	FGF-21 analogs	Placebo	FGF-21 analogs	Placebo	FGF-21 analogs	Placebo	FGF-21 analogs	Placebo
Efruxifermin (NCT03976401)	2021	IIa	16 weeks	Efruxifermin (n=59)	28mg QW (n=19)	Placebo (n=21)	50.4 (12.4)	52.4 (9.6)	10 (53)	15 (71)	19 (100)	19 (91)	38.8 (9.3)	37.6 (4.8)	7 (37)	14 (67)
					50mg QW (n=20)		52.6 (14.2)		10 (50)		18 (90)		36.7 (6.8)		10 (50)	
					70mg QW (n=20)		53.0 (13.2)		11 (55)		19 (95)		37.2 (5.5)		10 (50)	
Efruxifermin (NCT03976401)	2023	IIa	16 weeks	Efruxifermin (n=20)	50mg QW (n=20)	Placebo (n=10)	61.1 (10.0)	57.1 (14.4)	16 (80)	3 (30)	18 (90)	10 (100)	36.0 (5.6)	39.1 (8.2)	10 (50)	5 (50)
Efruxifermin (NCT04767529)	2023	IIb	24 weeks	Efruxifermin (n=85)	28mg QW (n=42)	Placebo (n=43)	57	55	29 (69)	27 (63)	NA	NA	NA	NA	32 (76)	28 (65)
					50mg QW (n=43)		52		23 (53)						30 (70)	
Pegbelfermin (NCT02413372)	2019	IIa	16 weeks	Pegbelfermin (n=45)	10mg q.d. (n=25)	Placebo (n=26)	52 (10)	47 (12)	15 (60)	16 (62)	24 (96)	25 (96)	34 (4)	37 (7)	9 (36)	11 (42)
					50mg QW (n=20)		52 (12)		17 (71)		23 (96)		35 (6)		8 (33)	
Pegbelfermin (NCT03486899)	2023	IIb	48 weeks	Pegbelfermin (n=148)	10mg QW (n=49)	Placebo (n=49)	56.4 (9.6)	57.5 (8.0)	29 (59)	29 (59)	43 (88)	41 (84)	36.3 (6.7)	35.2 (8.1)	34 (72)	34 (74)
					20mg QW (n=50)		56.3 (10.1)		27 (54)		42 (84)		35.1 (6.4)		35 (75)	
					40mg QW (n=49)		57.4 (10.5)		31 (63)		42 (86)		35.7 (6.6)		34 (74)	
Pegbelfermin (NCT03486912)	2023	IIb	48 weeks	Pegbelfermin (n=115)	10mg QW (n=39)	Placebo (n=39)	60.2 (8.0)	61.4 (7.5)	28 (72)	24 (61.5)	30 (77)	33 (85)	34.5 (5.9)	35.4 (6.4)	27 (77)	28 (78)
					20mg QW (n=37)		58.9 (9.3)		25 (68)		30 (81)		35.5 (6.2)		27 (79)	
					40mg QW (n=39)		57.0 (10.0)		21 (54)		34 (87)		36.9 (5.8)		28 (80)	
Pegozafermin (NCT04048135)	2023	IIb/IIa	12 weeks	Pegozafermin (n=62)	18/36mg QOW (n=62)	Placebo (n=19)	51.7 (10.0)	52.6 (9.0)	38 (61)	12 (63)	57 (92)	17 (89)	34.8 (5.2)	33.8 (2.8)	25 (40)	12 (63)
Pegozafermin (NCT04929483)	2023	IIb	24 weeks	Pegozafermin (n=151)	15mg QW (n=21)	Placebo (n=71)	55.0 (10.5)	56.3 (9.0)	9 (43)	39 (55)	18 (86)	67 (94)	37.8 (4.8)	38.1 (5.6)	18 (86)	49 (69)
					30mg QW (n=73)		55.3 (11.2)		50 (68)		69 (95)		35.1 (6.4)		45 (62)	
					44mg QOW (n=57)		55.2 (11.2)		37 (65)		54 (95)		36.1 (5.5)		35 (61)	

BMI, body mass index; FGF-21 analogs, fibroblast growth factor-21 analogs; n, number of patients; NA, not applicable; q.d., once per day; QW, once a week; QOW, every other week; SD, standard deviation; T2D, type 2 diabetes.

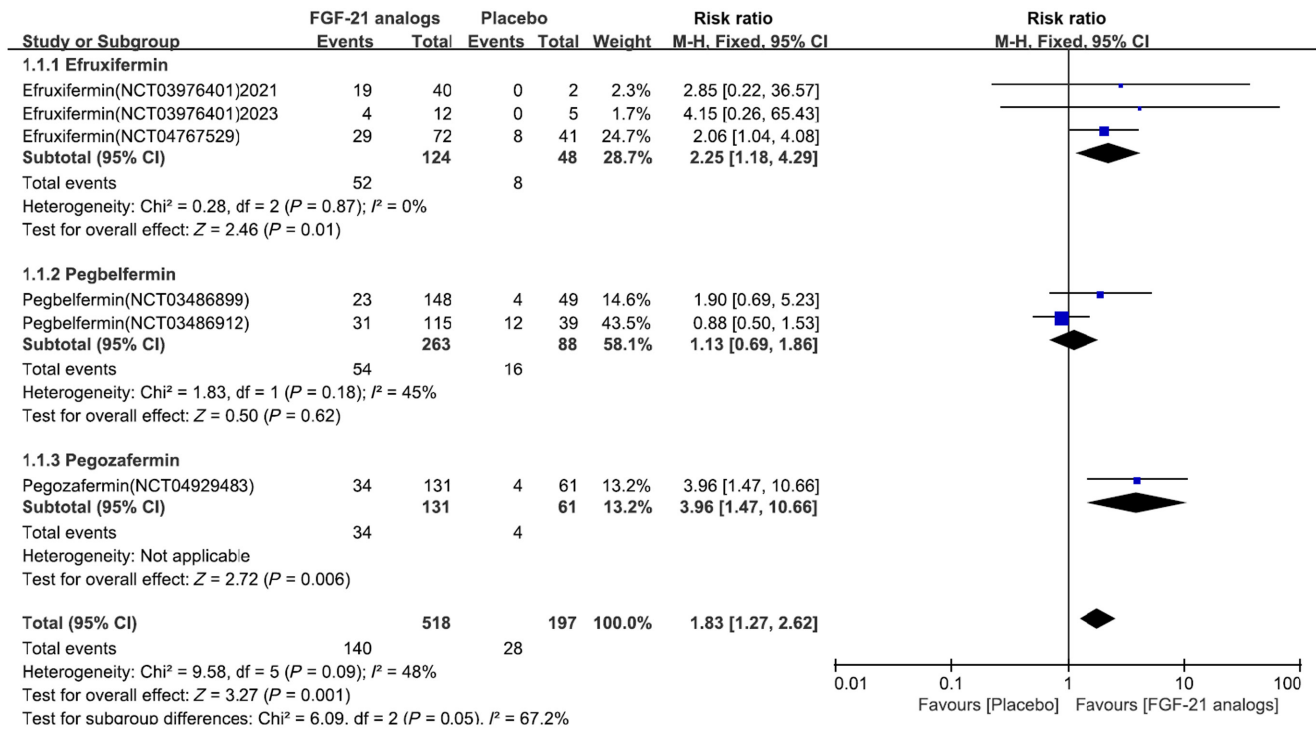


Figure 2 Forest plots of achieving a ≥ 1 stage improvement in fibrosis with no worsening of MASH in patients with MASLD/MASH in the FGF-21 analog and placebo groups.

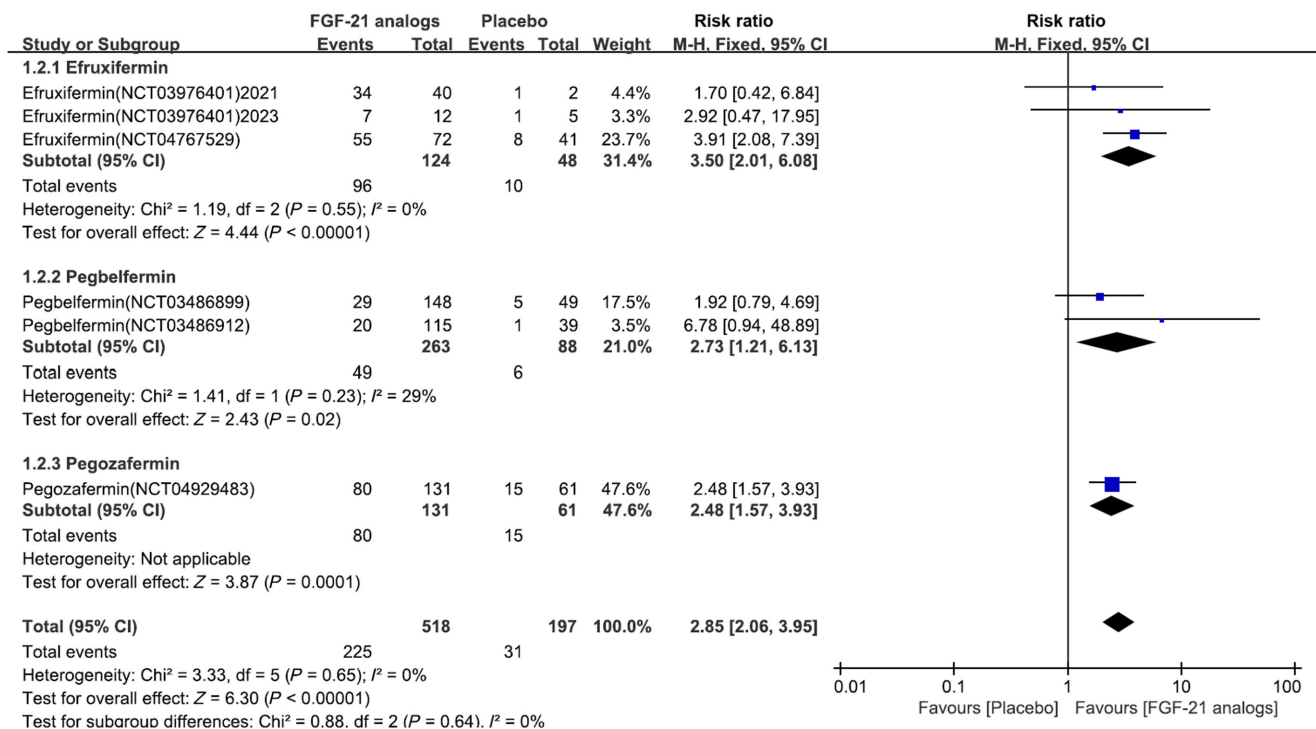


Figure 3 Forest plots of at least a two-point improvement in the MASLD activity score (NAS) with no worsening of fibrosis in patients with MASLD/MASH in the FGF-21 analog and placebo groups.

(RR = 3.50; 95% CI = 2.01–6.08; $P < 0.00001$) in achieving at least a two-point improvement in NAS with no worsening of fibrosis based on a single trial.

No significant differences were observed between the FGF-21 analogs and placebo in terms of MASH resolution with no worsening of fibrosis (RR = 2.52; 95% CI = 0.95–6.66; $P = 0.06$) with

moderate heterogeneity ($\chi^2 = 11.15$; $I^2 = 55\%$; $P = 0.05$) among the studies (Figure 4). Sensitivity analysis of the primary efficacy outcomes demonstrated that the meta-analysis was robust, and no significant publication bias was found according to Begg's test, Egger's test, and funnel plots (Figure S2, Table S3).

Liver fat content reduction

FGF-21 analogs demonstrated significant efficacy in achieving at least a 30% relative LFC reduction compared with the placebo group (Figure 5). The LFC reduction analysis included data from six RCTs involving 685 patients (493 receiving FGF-21 analogs

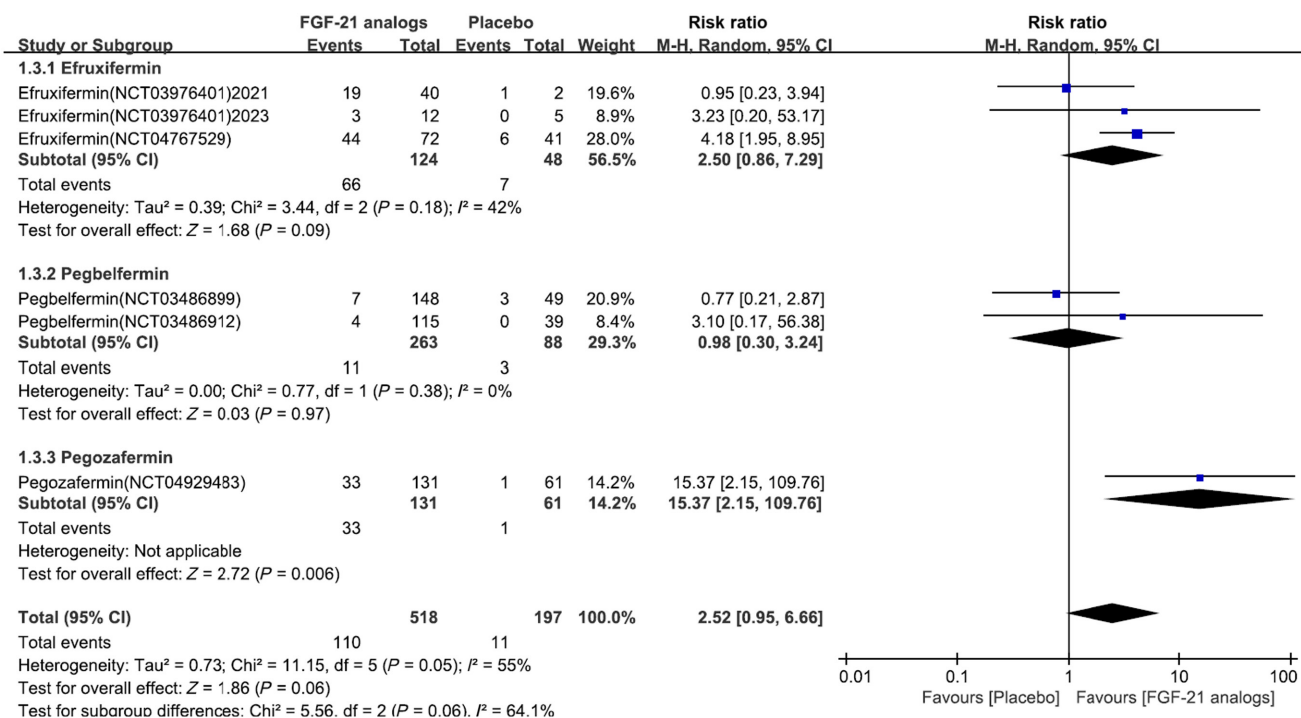


Figure 4 Forest plots of MASH resolution with no worsening of fibrosis in patients with MASLD/MASH in the FGF-21 analog and placebo groups.

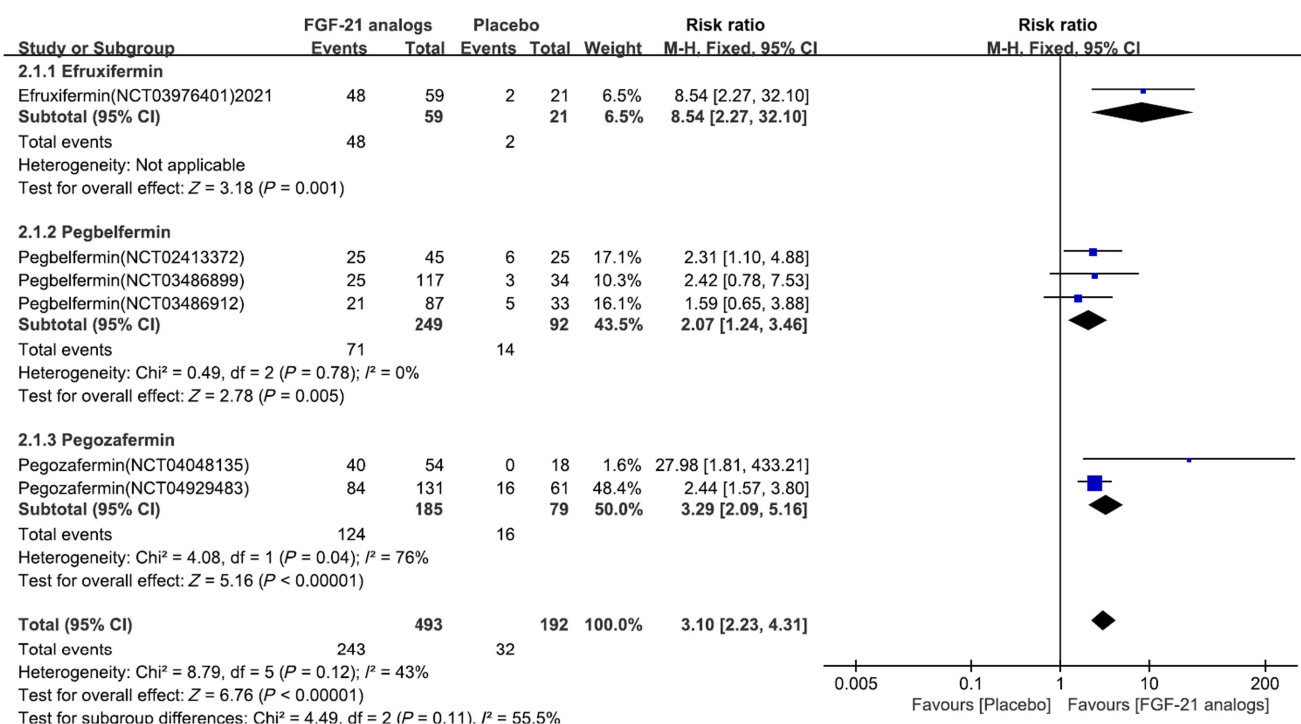


Figure 5 Forest plots of the liver fat content (LFC) reduction in patients with MASLD/MASH in the FGF-21 analog and placebo groups.

and 192 receiving a placebo). The pooled estimate for achieving at least a 30% relative LFC reduction was 3.10 (95% CI = 2.23–4.31; $P < 0.00001$), with moderate heterogeneity ($\chi^2 = 8.79$; $I^2 = 43\%$; $P = 0.12$) among the studies. In subgroup analysis, pegbelfermin showed the lowest efficacy for achieving LFC reduction (RR = 2.07; 95% CI = 1.24–3.46; $P = 0.005$), while efruxifermin exhibited the highest efficacy (RR = 8.54; 95% CI = 2.27–32.10; $P = 0.001$), even though it was based on a single study. Sensitivity analysis of the LFC reduction demonstrated that the meta-analysis was robust, and no significant publication bias was found according to Begg's test, Egger's test, and funnel plots (Figure S3, Table S3).

Biochemical outcomes

Results of various biochemical outcomes, including the percentage change from baseline for ALT, AST, TG, LDL-C, HDL-C, adiponectin, and Pro-C3, associated with exposure to FGF-21 analogs vs. placebo, are presented in Figure S4. Overall, the FGF-21 analogs demonstrated significant efficacy in achieving an improvement from baseline for ALT (MD = -18.79; 95% CI = -28.24 to -9.34; $P < 0.0001$), AST (MD = -20.60; 95% CI = -25.12 to -15.08; $P < 0.00001$), TG (MD = -15.56; 95% CI = -20.89 to -10.23; $P < 0.00001$), HDL-C (MD = 9.17; 95% CI = 3.68–14.65; $P = 0.001$), and adiponectin (MD = 20.33; 95% CI = 4.60–36.06; $P = 0.01$) compared with the placebo group.

In subgroup analysis, efruxifermin showed the highest efficacy for TG reduction (MD = -30.50; 95% CI = -47.67 to -13.33; $P = 0.0005$). While pegozafermin exhibited the highest efficacy for ALT (MD = -28.54; 95% CI = -37.33 to -19.75; $P < 0.00001$) and AST reduction (MD = -24.20; 95% CI = -32.15 to -16.24; $P < 0.00001$), which are typical markers of liver function and injury. Additionally, pegozafermin demonstrated the highest efficacy for HDL-C (MD = 13.34; 95% CI = 9.64–17.03; $P < 0.00001$) and adiponectin increase (MD = 35.38; 95% CI = 24.47–46.29; $P < 0.00001$), even though efruxifermin could not be included. Sensitivity analysis of the biochemical outcomes demonstrated that the meta-analysis was robust, and no significant publication bias was found according to Begg's test, Egger's test, and funnel plots (Figure S5, Table S3).

Safety outcomes

The safety outcomes (TEAEs, treatment-related TEAEs, TEAEs leading to discontinuation, serious TEAEs, injection site erythema, diarrhea, and nausea) associated with exposure to FGF-21 analogs vs. placebo are presented in Figure S6. Overall, FGF-21 analogs were significantly associated with an increased risk of the incidence of TEAEs (RR = 1.17; 95% CI = 1.08–1.27; $P = 0.0002$), treatment-related TEAEs (RR = 1.75; 95% CI = 1.40–2.19; $P < 0.00001$), injection site erythema (RR = 2.33; 95% CI = 1.29–4.20; $P = 0.005$), diarrhea (RR = 2.00; 95% CI = 1.42–2.81; $P < 0.0001$), and nausea (RR = 2.01; 95% CI = 1.41–2.85; $P = 0.0001$).

In subgroup analysis, efruxifermin showed the highest risk of the incidence of TEAEs (RR = 1.39; 95% CI = 1.15–1.67;

$P = 0.0006$), treatment-related TEAEs (RR = 2.28; 95% CI = 1.35–3.87; $P = 0.002$), diarrhea (RR = 2.52; 95% CI = 1.41–4.49; $P = 0.002$), and nausea (RR = 3.33; 95% CI = 1.65–6.69; $P = 0.0008$). Pegozafermin exhibited the highest risk of the incidence injection site erythema (RR = 2.61; 95% CI = 0.79–8.60; $P = 0.12$). Sensitivity analysis of the safety outcomes demonstrated that the meta-analysis was robust, and no significant publication bias was found according to Begg's test, Egger's test, and funnel plots (Figure S7, Table S3).

DISCUSSION

This systematic review and meta-analysis comprehensively evaluated the clinical efficacy and safety of newly introduced treatments for MASLD/MASH, specifically focusing on medications that target FGF-21. Pooled data from eight RCTs involving FGF-21 analogs (efruxifermin, pegbelfermin, and pegozafermin) demonstrated that these agents were significantly more effective compared with placebo in terms of several relevant primary and secondary outcomes. Despite the increased risk of TEAEs and treatment-related TEAEs with FGF-21 analogs, there was no significant difference in the incidence of TEAEs leading to discontinuation or serious TEAEs between the FGF-21 analogs and placebo. This is the first systematic review and meta-analysis to comprehensively demonstrate the efficacy and safety of FGF-21 analogs in patients with MASLD/MASH.

According to the current regulatory guidelines, liver histology assessment, including steatosis, inflammation, ballooned hepatocytes, and fibrosis, serves as a composite for study entry and as a surrogate efficacy end point in MASH phase IIb and phase III clinical trials.^{25,26} Based on these guidelines, the primary outcomes to demonstrate the efficacy of FGF-21 analogs were set as liver histological improvements, including achieving a ≥ 1 stage improvement in fibrosis with no worsening of MASH, at least a two-point improvement in the NAS with no worsening of fibrosis, and MASH resolution with no worsening of fibrosis. The relationship between fibrosis improvement and disease outcomes in MASH patients is well-established.²⁷ Additionally, recent study has demonstrated a correlation between MASH resolution and fibrosis improvement,²⁸ highlighting the significance of MASH resolution as an end point due to its role in driving fibrosis development.²⁹

FGF-21 analogs were found to be significantly more effective in achieving a ≥ 1 stage improvement in fibrosis with no worsening of MASH and at least a two-point improvement in the NAS with no worsening of fibrosis compared with placebo. However, they did not exhibit significant effectiveness in terms of MASH resolution or the worsening of fibrosis. MASH resolution is considered more stringent than other outcomes as it requires a ballooning score of 0 and an inflammation score of 0/1. Further studies with larger sample sizes and longer durations are required to provide a more definitive understanding of the effects of FGF-21 analogs on MASH resolution.

A previous study suggested that FGF-21 analogs have the potential to treat signs of metabolic syndrome, such as high blood insulin levels, obesity, and hypercholesterolemia.³⁰ Consistent with this report, FGF-21 analogs significantly decreased TG levels by more

than 15.56% and increased HDL-C levels by more than 9.1% compared with placebo in this study. Adiponectin, which promotes insulin sensitization, fat burning, and exerts anti-inflammatory and antioxidant effects,³¹ was found to be significantly increased by more than 20.03% by use of FGF-21 analogs in MASLD/MASH patients compared with placebo. This indicates that FGF-21 analogs can effectively improve the metabolic risk factors involved in MASLD/MASH development and progression compared with placebo. Although there was no significant decrease in LDL-C concentration between the FGF-21 analogs and placebo groups, the positive impact of FGF-21 analogs on lipid profiles may be considered a strength compared with obeticholic acid, another potential treatment for MASLD/MASH that has adverse effects on lipid profiles.³²

While FGF-21 analogs have a similar molecular structure to endogenous FGF-21, they have been engineered to have an increased half-life and extended activity. Efruxifermin consists of two modified human FGF-21 variants linked by short polyglycine-serine sequences to a human immunoglobulin G1 Fc domain.³³ In contrast, pegbelfermin is a polyethylene glycol-modified (PEGylated) recombinant human FGF-21 analog.³⁴ Pegzofermin, a more recent development, is a glycoPEGylated FGF-21 analog with a prolonged half-life that surpasses that of efruxifermin and pegbelfermin.³⁵ Our meta-analysis demonstrated that pegbelfermin exhibited the lowest effectiveness overall in terms of liver histological improvement and biochemical outcomes among the three FGF-21 analogs. This finding supports manufacturers' decision to no longer focus on pegbelfermin development.¹⁴ Pegzofermin demonstrated the highest effectiveness in improving liver histology and biochemical outcomes. Although we could not perform a meta-analysis according to dose owing to the lack of a sufficient number of RCTs for pegzofermin, results from a study by Loomba *et al.*²⁴ indicated potential dose-dependent improvements in fibrosis, with 22%, 26%, and 27% of patients in the 15, 30, and 44 mg pegzofermin groups, respectively, compared with 7% in the placebo group. Additionally, pegzofermin induced MASH resolution without worsening fibrosis, with remarkable differences compared with placebo in the 15 mg (35%), 30 mg (21%), and 44 mg (24%) groups. Further studies are needed to explore the clinical efficacy of FGF-21 analogs based on various dosages in patients with MASLD/MASH as more data are available from future RCTs.

In this study, the FGF-21 analogs demonstrated an acceptable safety profile. Injection site erythema and gastrointestinal conditions were the most prevalent side effects. Importantly, these side effects did not escalate to serious TEAEs or necessitate treatment discontinuation. It is noteworthy that patients with MASLD/MASH who received obeticholic acid had an increased risk of pruritus, a condition requiring management strategies, such as the use of antipruritic medications or additional pharmacotherapy.³² The potential contrasting safety profiles between FGF-21 analogs and obeticholic acid underscore the importance of considering not only efficacy but also tolerability and side-effect profiles when evaluating treatment options for MASLD/MASH. Further studies directly comparing FGF-21 analogs and obeticholic acid may provide valuable insights for clinicians to select the most suitable treatment for patients with MASLD/MASH.

Despite the novelty of our meta-analysis, it has several limitations. First, the variability in the number of RCTs associated with each active treatment, and the low number of events reported in some trials, may have limited the robustness of certain outcomes. In certain cases, the data were based on only one RCT, which hindered definitive conclusions. Second, some of the included RCTs had relatively small sample sizes, which may have affected the statistical power and precision of the results. Third, the limited number of clinical trials (only eight studies) included in this meta-analysis may have affected the generalizability of the findings. However, it is important to note that all trials were of high quality and double-blind, randomized, and placebo-controlled studies.

In conclusion, this systematic review and meta-analysis provides comprehensive insights into the clinical efficacy and safety of FGF-21 analogs for the treatment of MASLD/MASH. Pooled data from eight RCTs revealed that the FGF-21 analogs, efruxifermin, pegbelfermin, and pegzofermin, were significantly better in achieving liver histological improvements and beneficial biochemical outcomes compared with placebo. Despite the increased risk of TEAEs and treatment-related TEAEs, FGF-21 analogs demonstrated an acceptable safety profile, with the most prevalent side effects being injection site erythema and gastrointestinal conditions, which did not lead to serious TEAEs or treatment discontinuation. Notably, variations were observed in the efficacy of different FGF-21 analogs, with pegzofermin showing the highest efficacy among all tested candidates. These findings shed light on the efficacy and safety of FGF-21 analogs and provide valuable evidence for their application as MASLD/MASH therapeutics.

SUPPORTING INFORMATION

Supplementary information accompanies this paper on the *Clinical Pharmacology & Therapeutics* website (www.cpt-journal.com).

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CONFLICT OF INTEREST

The authors declared no competing interests for this work.

AUTHOR CONTRIBUTIONS

C.J., N.J., C.E.S., and I.B. wrote the manuscript. C.J., N.H., and I.B. designed the research. C.J., N.H., N.J., S.R., C.E.S., M.S.K., and I.B. performed the research. C.J., N.H., N.J., S.R., C.E.S., M.S.K., and I.B. analyzed the data.

ETHICS APPROVAL

As this is a secondary literature-based study, ethic approval is not necessary.

DATA AVAILABILITY STATEMENT

The data sets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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